

- 6 Fertilisation in mammals, such as the African northern white rhino, results in the fusion of gamete nuclei.

The photograph shows the last two remaining northern white rhinos.

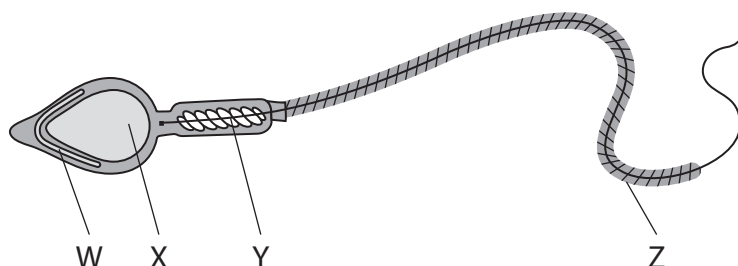
Both of these rhinos are female.



(Source: © Xinhua/Alamy Stock Photo)

Sperm and tissue samples were taken from the last few male northern white rhinos before they died. The sperm samples were frozen.

- (a) The photograph shows an organelle taken from a sperm cell, as seen using an electron microscope. The diagram shows a sperm cell.



(Source: © Science History Images/Alamy Stock Photo)

In which part of this sperm cell would this organelle be found?

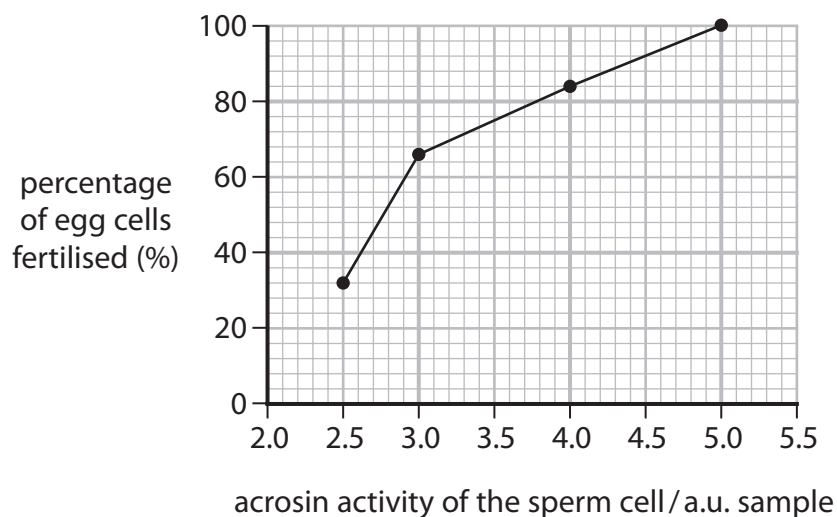
(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(b) Sperm cells contain a digestive enzyme called acrosin.

The effect of acrosin activity of samples of sperm cells on the percentage of egg cells fertilised was investigated.

The graph shows the results of this investigation.



(i) Describe the role of **one** organelle involved in the production of this enzyme.

(2)

Organelle

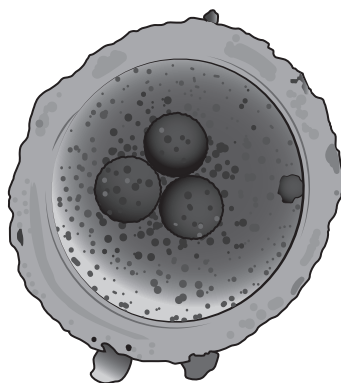
Role



- (c) In vitro fertilisation (IVF) has been used to fertilise northern white rhino egg cells using the frozen sperm samples.

The zona pellucida of some egg cells were damaged when they were removed from the ovaries of the females.

The drawing shows a fertilised egg cell with three nuclei.



magnification $\times 200$

- (i) State how to calculate the actual size of this egg cell.

(1)

- (ii) Suggest why this fertilised egg cell has three nuclei.

(3)

(Total for Question 6 = 12 marks)

